

iKWATH

UI / UX DESIGN HANDOFF

Complete design specification for the Raspberry Pi touchscreen, research interface and companion web/mobile UI.

Goal: Give the UI designer enough detail to design the complete product without guessing screens, states, flows or required data.

Project	iKwath
SIH Problem	SIH26048 — iKwath
Primary computer	Raspberry Pi
Real-time layer	ESP32 + sensors/actuators
Core UX	Simple for users; transparent for engineering/research

VERSION 1.0 • DESIGN TEAM HANDOFF

1. Product Vision

iKwath should look like a premium scientific appliance, not a generic smart kettle. The UI must communicate that each pod loads a formulation-specific process and that the machine measures the extraction rather than simply running a timer.

Core product statement

Insert the validated pod → iKwath loads its recipe → the machine executes and measures the extraction → the brew is recorded → the system cleans itself.

Three UX layers

Layer	User	Purpose
User Mode	Home/clinic operator	One simple flow; minimal decisions
Technician Mode	Service/operator	Diagnostics, calibration and actuator tests
Research Mode	Team/lab	Graphs, experiments, validation and export

Design personality

- Clean, calm, trustworthy and technical.
- Large numbers for temperature, mass and progress.
- Subtle botanical cues; avoid excessive Ayurveda decoration.
- Use icons plus text for critical states.
- Do not use AI/medical claims unless the team has evidence.

2. Information Architecture

Recommended navigation for the Pi touchscreen and companion browser UI.

Area	Content
Home	Machine readiness, pod status, start brew, last brew
Brew	Live stages, temperature, mass, progress, pause/cancel
History	Brew records, search, filters, details, export
Formulations	Validated profiles; read-only for normal users
Validation	Traditional vs iKwath comparison, graphs, metrics
Cleaning	Rinse/drain state and cleaning history
System	Connectivity, sensors, calibration, maintenance
Research	Analytics and experimental data

Main user flow

HOME → INSERT POD → POD DETECTED → PROFILE LOADED → CONFIRM → LIVE BREW → FILTRATION → DISPENSE → BREW PASSPORT → CLEANING → READY

Designer must create reusable components

- Buttons, cards, metric tiles, progress/stage stepper, gauges.
- Pod status, recipe parameter rows, alerts and confirmation dialogs.
- Graphs, tables, status chips, pump/heater/valve indicators.
- Loading, empty, offline, warning, fault and success states.

3. Home Screen

The home screen answers: Is the machine ready? What should I do? Is anything blocking me?



States

State	Message
READY	Ready for a new brew.
BUSY	Brew in progress.
CLEANING	Cleaning cycle in progress.
ATTENTION	Action required before starting.
OFFLINE	Local machine available; network unavailable.

Rule: the primary action is always obvious. Do not put graphs or engineering data on this screen.

4. Pod Detection & Formulation

The pod is the physical key that maps to a machine-executable formulation profile.

Flow

INSERT POD → READING POD... → POD DETECTED → PROFILE LOADED → CONFIRM

UI field	Example
Formulation	Ashwagandha Kwatha
Pod ID	AKW-000421
Profile	Validated formulation profile
Water	Profile-defined
Temperature	Profile-defined
Soaking	Profile-defined
Stirring	Profile-defined
Reduction endpoint	Profile-defined

Important

Do not hard-code one universal recipe. The SIH example uses approximately 400 mL input and 100 mL output, but the machine should use the formulation-specific profile where applicable.

Failure states

- Read failed → “Pod could not be read. Remove and reinsert.”
- Invalid/unrecognized pod → “This pod is not supported.”
- Profile unavailable → block brew and explain that a validated profile is required.
- Normal users cannot edit validated parameters.

7. Extraction Journey & Stage States

Stage	User label	Backend concept
1	Pod detected	POD_DETECTED
2	Soaking	SOAKING
3	Extraction	EXTRACTION
4	Controlled reduction	REDUCTION
5	Filtration	FILTRATION
6	Dispensing	DISPENSING
7	Cleaning	CLEANING
8	Ready	READY

Animation

- Completed stages become checkmarks.
- Current stage is emphasized; future stages are muted.
- Use subtle movement only; never distract during heating or dispensing.
- A paused process must show PAUSED plus the reason.

Example microcopy

- Soaking — “Preparing the powder for extraction.”
- Extraction — “Maintaining the formulation profile.”
- Reduction — “Concentrating toward the target endpoint.”
- Filtration — “Separating spent coarse powder.”
- Dispensing — “Fresh decoction is ready to dispense.”

8. Adaptive Reduction UI

The software should visually prove that iKwath is measurement-driven rather than simply timer-driven.



Concept

Temperature + mass + time → reduction controller → endpoint assessment → continue / stabilize / stop

Research detail screen

- Mass vs time graph
- Temperature vs time graph
- Current mass + target
- Reduction rate/trend
- Stage transition timestamps
- Final mass/volume where reliably measured
- Optional extract density
- Total cycle time

Important scientific UX rule

Do not claim that the UI or algorithm itself proves medicinal equivalence. Experimental comparison with a reference preparation is required before making such a claim.

9. Filtration & Dispensing

The physical concept uses bottom extraction from the bowl, an external removable filter, a pump and dispensing path. The UI should show this as a clean closed process.

REDUCTION COMPLETE ↓ FILTERING ✓ Product path selected ● Pump active ● Filter path active ↓ DISPENSING
✓ Filtered liquid ● Output active ↓ BREW READY

Suggested screen

- “Filtering spent coarse powder.”
- “Dispensing fresh decoction.”
- Show pump/valve activity only when relevant.
- If flow is blocked, stop safely and provide an actionable service message.

Do not show “filtered” unless the system actually completed the filtration step.

11. Automatic Cleaning UI

Cleaning should be treated as a first-class product state. The next brew can remain locked until the required cleaning cycle completes.

CLEANING CYCLE ✓ Product path closed ● Rinse water flowing ■ Internal path flush ■ Drain ■ System reset 62% "Keep the dispensing area clear."

State	Message
Required	Cleaning required before next brew.
Rinsing	Rinse water flowing.
Draining	Draining the process path.
Complete	Cleaning complete — system ready.
Failed	Cleaning incomplete — check drain path.

Design rule

Do not unlock the next brew after an incomplete cleaning cycle unless a technician deliberately overrides it.

12. History & Search

SEARCH: [Ashwagandha] FILTER: [PASS ▼] #421 Ashwagandha 101.8 g 18:42 ✓ #420 Dashamoola 100.4 g 21:10 ✓ #419 Ashwagandha 102.1 g 18:35 ✓ [VIEW] [COMPARE] [EXPORT]

Filters

- Formulation
- Date range
- Pass / warning / failed
- Cycle duration
- Endpoint deviation
- Validation runs

History detail

- Open Brew Passport.
- Show process graph and key measurements.
- Show warnings/events.
- Allow export from Research/Technician Mode.

13. Traditional vs iKwath Validation Mode

This is the strongest research-facing screen. It demonstrates that the team is measuring the core challenge: reducing practical preparation time while comparing the resulting process/output with a reference preparation.

Metric	Traditional	iKwath	UI
Initial water	Measured	Measured	Compare
Temperature profile	Recorded	Recorded	Overlay graphs
Final mass	Measured	Measured	Compare
Final volume	Measured	Measured if available	Compare
Extract density	If measured	If measured	Compare
Cycle time	Measured	Measured	Compare
Marker constituents	Lab data if available	Lab data if available	Scientific result

Graph layout

Use two overlaid curves with a clear legend: TRADITIONAL vs iKWATH. Provide zoom, tooltip and run selection in the desktop/research version.

Never invent pass/fail thresholds. Acceptance criteria should come from the team's experimental protocol and relevant reference method.

14. Technician Mode

Technician Mode is for diagnostics, maintenance and controlled actuator tests.

Subsystem	Example states
Temperature sensor	CONNECTED / FAULT
Load cell	CONNECTED / CALIBRATION REQUIRED
Heater	OFF / ACTIVE / FAULT
Stirrer	OFF / ACTIVE / FAULT
Pump	OFF / ACTIVE / FAULT
Product valve	OPEN / CLOSED
Drain valve	OPEN / CLOSED
Pod reader	CONNECTED
Raspberry Pi	ONLINE
ESP32	CONNECTED / LOST

Actions

- Sensor diagnostics
- Load-cell calibration
- Temperature calibration
- Pump prime/test
- Valve test
- Stirrer test
- Heater test with interlock
- Rinse test
- Error log
- Controller restart

Actuator tests must require deliberate confirmation and safety checks.

15. Research Mode & Analytics

Research Mode turns the prototype into a measurement and evidence platform.

Dashboard cards

Card	Purpose
Endpoint consistency	Compare repeated final endpoints
Cycle time	Measure practical preparation time
Temperature stability	Assess process control
Brew repeatability	Compare repeated runs
Cleaning compliance	Track completed cleaning cycles

Data shown

- Run ID, formulation, pod ID, timestamps
- Temperature curve
- Mass curve
- Stirring state/RPM if measured
- Pump/valve events
- Endpoint timestamp
- Final measured values
- Warnings/faults

Export formats

- CSV for experiments
- PDF/printable brew report
- JSON/API if a web dashboard is later connected

16. Alerts, Errors & Recovery

Condition	User message	Action
No pod	Insert a validated pod.	Wait
Read failed	Pod could not be read.	Retry
Low water	Add water before starting.	Block start
Temperature fault	Temperature sensor unavailable.	Safe stop
Over-temperature	Temperature outside safe range.	Stop heater
Dry-run risk	Insufficient liquid detected.	Stop heating
Filter blockage	Flow restriction detected.	Stop pump/service
Pump fault	Dispensing path needs attention.	Stop pump
ESP32 disconnected	Controller connection lost.	Safe stop
Cleaning failed	Cleaning incomplete.	Lock next brew

Severity labels

- INFO — information only.
- WARNING — attention needed.
- STOP — machine safely paused/stopped.
- SERVICE — technician intervention required.

Never write “Something went wrong.” Say what happened and what to do.

18. Formulation Library & Versioning

ASHWAGANDHA KWATHA Validated profile Water Profile-defined Temperature Profile-defined Soaking
Profile-defined Stirring Profile-defined Reduction endpoint Profile-defined [VIEW PROFILE]

Profile sections

- Identity — formulation name, profile ID and revision.
- Preparation — machine-executable parameters.
- Validation — status and reference run.
- Usage — number of runs.
- Revision history — changes and approval.

Versioning

PROFILE v1.0 → validated → PROFILE v1.1 → revalidated → ACTIVE PROFILE

Normal users must not silently edit validated parameters. A change should create a new revision and enter the research/validation workflow.

19. Mobile / Web Companion

A companion interface can expose history, machine status and research data without making remote control the focus of the first prototype.

```
iKWATH ● Machine Online Ashwagandha Kwatha Ready to brew Last Brew #421 • PASS [ HISTORY ] [ VALIDATION ] [ MACHINE ]
```

Recommended features

- Brew history and Brew Passport
- Validation reports
- Machine status
- Maintenance reminders
- Read-only formulation library
- Data export

Safety

Avoid remote-start in the first prototype unless the physical safety model is fully defined. A remote tap should not unexpectedly start heating a physical appliance.

20. Design System

The designer should create a component library before producing the final screens.

Category	Requirement
Typography	Legible sans-serif; strong hierarchy; large process numbers
Buttons	Large touch targets; primary/secondary/destructive variants
Status	Text + icon + semantic color; never color alone
Cards	Consistent radius, padding and hierarchy
Icons	One coherent icon family
Graphs	Readable labels, legends, tooltips
Touchscreen	Finger-friendly spacing and controls
Responsive	Pi touchscreen + laptop/browser + mobile

Recommended semantic colors

- Success — ready/completed/passed
- Warning — attention/deviation
- Danger — stop/fault
- Neutral — inactive/information
- Accent — primary interaction/current stage

Keep the visual language premium and restrained. Avoid making every component green simply because the product is Ayurvedic.

21. Required UI States

Screen	States to design
Home	Ready / Busy / Cleaning / Attention / Offline
Pod	Scanning / Detected / Invalid / Read failed
Start	Ready / Missing water / Not clean / Unsafe
Brew	All stages / Paused / Cancel confirm / Fault
Temperature	Normal / Warning / Sensor fault
Mass	Normal / Unexpected trend / Sensor fault
Dispense	Ready / Active / Blocked / Complete
Cleaning	Required / Rinsing / Draining / Failed / Complete
History	Data / Empty / Loading / Exporting
Validation	No data / Comparison / Export
Technician	Healthy / Fault / Calibration

Figma rule

Do not deliver only the happy path. Judges, developers and testers need the abnormal states too.

22. UI ↔ Software Data Contract

The designer does not need to code the backend, but the UI should be based on data the Raspberry Pi/ESP32 system can actually provide.

```
{ brew_id: "421", pod_id: "AKW-000421", formulation: "Ashwagandha Kwatha", stage: "REDUCTION",  
  temperature_c: 87.6, mass_g: 168.0, target_mass_g: 105.0, stirrer: "ACTIVE", heater: "ACTIVE", pump:  
  "OFF", cleaning_required: false, status: "RUNNING" }
```

Useful events

- pod_detected
- profile_loaded
- water_ready
- soaking_started/completed
- extraction_started
- reduction_started
- endpoint_reached
- filtration_started/completed
- dispensing_started/completed
- cleaning_started/completed
- warning
- fault
- brew_completed

State-driven UI

Animations should follow real machine state/events rather than fake timers. This makes the demo credible and prevents the screen from saying “Filtering” while the hardware is doing something else.

23. What We Should Call Unique

Feature	Positioning
Pod-based preparation	SIH requirement — not unique alone
Formulation profile	SIH requirement — not unique alone
Controlled temperature/reduction	SIH requirement — not unique alone
Weight/density monitoring	SIH requirement — not unique alone
Automatic cleaning	SIH requirement — not unique alone
Raspberry Pi	Implementation choice
NFC	Implementation choice
Adaptive endpoint visualization	Strong software/control differentiator
Brew Passport	Traceability differentiator
Traditional vs iKwath validation mode	Research/demo differentiator
User + Technician + Research modes	System UX differentiator
State-driven closed-loop UI	Engineering UX differentiator

Presentation rule

Do not claim “world's first”, “AI-powered”, “100% standardized” or similar statements without evidence. Your advantage is the integrated engineering implementation and validation workflow.

24. SIH Demo Flow

The UI should support this exact demonstration story:

1. HOME – “System ready.” 2. INSERT POD – automatic detection. 3. PROFILE – formulation + loaded recipe. 4. START – one tap. 5. LIVE BREW – temperature, mass and stage. 6. ADAPTIVE REDUCTION – endpoint trend. 7. FILTRATION – closed-path filtration. 8. DISPENSE – fresh warm decoction. 9. CLEAN – automatic rinse/drain. 10. BREW PASSPORT – recorded result. 11. VALIDATION – Traditional vs iKwath comparison.

The judge should understand the system from the screens even before the presenter explains the engineering.

25. Exact Figma Deliverables

Figma page	Frames
00 — Design System	Logo, typography, colors, icons, components
01 — User Flow	Happy path + error branches
02 — Home	All machine states
03 — Pod	Scan/detected/invalid/failure
04 — Brew	Confirm/live/reduction/pause/cancel
05 — Completion	Dispense/ready/passport
06 — Cleaning	Required/rinse/drain/fail/complete
07 — History	List/detail/filter/empty/export
08 — Validation	Comparison/graphs/results/export
09 — Technician	Diagnostics/calibration/tests
10 — Research	Analytics/raw data
11 — Mobile/Web	Responsive companion
12 — Prototype	Clickable end-to-end demo

For every major screen include

- Normal state
- Loading state
- Success state
- Warning state
- Error state
- Empty state where applicable
- Interaction notes
- Responsive/touchscreen behavior

26. Final UX Blueprint

INSERT POD ↓ IDENTIFY FORMULATION ↓ LOAD VALIDATED RECIPE ↓ ADD WATER ↓ START ↓ MEASURE + CONTROL
EXTRACTION ↓ ADAPT REDUCTION ↓ FILTER + DISPENSE ↓ SAVE BREW PASSPORT ↓ RINSE + DRAIN ↓ SYSTEM READY

The three screens that deserve the most design effort

- **Live Brew** — the hero product screen.
- **Validation Mode** — the strongest technical/research screen.
- **Brew Passport** — the strongest traceability/software screen.

Final message to the UI designer

Do not design iKwath as a smart kettle. Design it as a formulation-aware, measurement-driven extraction system with a simple consumer flow and a deep research/engineering layer underneath.

END OF DESIGN HANDOFF